Academic Records Validator Using Blockchain Technology

A Project Report Submitted in the
Partial Fulfillment of the Requirements
for the Award of the Degree of

BACHELOR OF TECHNOLOGY

IN

COMPUTER SCIENCE AND ENGINEERING

Submitted by

G.Yashaswini	**23881A05FH**
K.Trivikram	**23881A05FU**
N.Laalu Naik	**23881A05GF**

SUPERVISOR
Dr.V.Mohan
Assistant Professor

VARDHAMAN
COLLEGE OF ENGINEERING

Department of Computer Science and Engineering

April, 2026

Department of Computer Science and Engineering

CERTIFICATE

This is to certify that the project titled **Academic Records validator Using Blockchain Technology** is carried out by

G.Yashaswini	**23881A05FH**
K.Trivikram	**23881A05FU**
N.Laalu Naik	**23881A05GF**

in partial fulfillment of the requirements for the award of the degree of **Bachelor of Technology** in **Computer Science and Engineering** during the year 2025-26.

Signature of the Supervisor
Dr.V.Mohan
Assistant Professor

Signature of the HOD
Dr. Gagandeep Arora
Professor and Head, CSE

Project Viva-Voce held on ______________________

Examiner

Kacharam (V), Shamshabad (M), Ranga Reddy (Dist.)–501218, Hyderabad, T.S.
Ph: 08413-253335, 253201, Fax: 08413-253482, www.vardhaman.org

Acknowledgements

The satisfaction that accompanies the successful completion of the task would be put incomplete without the mention of the people who made it possible, whose constant guidance and encouragement crown all the efforts with success.

We wish to express our deep sense of gratitude to **Dr. V. Mohan**, Assistant Professor and Project Supervisor, Department of Computer Science and Engineering, Vardhaman College of Engineering, for his able guidance and valuable suggestions, which helped us in completing the project on time.

We are particularly thankful to **Dr. Gagandeep Arora**, the Head of the Department, Department of Computer Science and Engineering, his guidance, intense support and encouragement, which helped us to mould our project into a successful one.

We show gratitude to our honorable Principal **Dr. J.V.R. Ravindra**, for providing all facilities and support.

We avail this opportunity to express our deep sense of gratitude and heartful thanks to **Dr. Teegala Vijender Reddy**, Chairman and **Sri Teegala Upender Reddy**, Secretary of VCE, for providing a congenial atmosphere to complete this project successfully.

We also thank all the staff members of Computer Science and Engineering department for their valuable support and generous advice. Finally thanks to all our friends and family members for their continuous support and enthusiastic help.

G.Yashaswini

K.Triviram

N.Laalu Naik

Abstract

In the digital age, academic certificates play a crucial role in employment, higher education, and identity verification. However, the increasing prevalence of forged and tampered certificates has become a major concern for institutions, organizations, and verification agencies. Traditional verification methods are often manual, time-consuming, and prone to human errors, making them inefficient and unreliable. To address these challenges, this project proposes an **Academic Records Validator**, a secure and automated system designed to verify the authenticity of academic certificates using advanced technologies.

The proposed system integrates **Optical Character Recognition (OCR), Artificial Intelligence (AI), and Blockchain technology** to ensure accurate, fast, and tamper-proof verification. Initially, the user uploads academic certificates such as 10th, Intermediate, and Degree documents. The system preprocesses the certificate images using image processing techniques like grayscale conversion, noise removal, and enhancement through OpenCV to improve quality. Then, Tesseract OCR is used to extract important details such as student name, roll number, certificate ID, institution name, year of passing, and marks.

To enhance accuracy, a hybrid data extraction method is employed to structure the extracted information into meaningful fields. The system further incorporates an AI-based forgery detection module using the MobileNetV2 deep learning model. This model analyzes the visual characteristics of the certificate and identifies inconsistencies or signs of manipulation, helping to detect fake documents effectively.

For ensuring data integrity and security, the extracted certificate data is converted into a unique digital fingerprint using the **SHA-256 hashing algorithm**. Instead of storing the entire certificate, only the generated hash is stored on the Ethereum blockchain using smart contracts developed in Solidity. This ensures immutability, transparency, and protection against data tampering.

During the verification process, the certificate is reprocessed, and a new hash is generated from the extracted data. This hash is then compared with the hash stored on the blockchain. If both hashes match, the certificate is verified as genuine; otherwise, it is flagged as fake. The final decision is strengthened by combining both hash comparison and AI-based forgery detection results.

The system provides a fast, reliable, and scalable solution for verifying multiple academic records within a single platform. It significantly reduces manual effort, minimizes errors, and enhances trust in certificate validation processes. The proposed solution can be effectively used by universities, employers, and government organizations for secure and efficient verification.

In conclusion, the Academic Records Validator demonstrates how the integration of AI and blockchain technologies can revolutionize the process of academic credential verification, providing a robust and future-ready solution to combat certificate fraud.

Keywords: Academic Records Validator, Certificate Verification, Optical Character Recognition (OCR), Artificial Intelligence (AI), MobileNetV2, Blockchain, Ethereum, SHA-256 Hashing, Forgery Detection, Smart Contracts

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Abbreviations

0.1 Abbreviations

- **AI** – Artificial Intelligence
- **ML** – Machine Learning
- **CNN** – Convolutional Neural Network
- **SHA-256** – Secure Hash Algorithm 256-bit
- **API** – Application Programming Interface
- **UI** – User Interface
- **GPU** – Graphics Processing Unit
- **PDF** – Portable Document Format
- **ETH** – Ethereum
- **SC** – Smart Contract
- **OCR** – Optical Character Recognition

CHAPTER 1

Introduction

1.1 Introduction

In the digital era, academic certificates play a vital role in employment, higher education, and identity verification. However, the increasing number of forged and tampered certificates has become a major concern for institutions, employers, and verification agencies. Traditional verification methods are mostly manual, time-consuming, and prone to human errors, making them inefficient and unreliable.

Brief Description of the Project:

The Academic Records Validator is a secure and automated system designed to verify academic certificates and prevent the use of fake or tampered documents. It integrates Optical Character Recognition (OCR), Artificial Intelligence (AI), and Blockchain to provide a fast, accurate, and reliable verification process.

The system allows users to upload certificates, which are preprocessed and analyzed using OCR to extract important details. An AI model (MobileNetV2) detects forgery, while a SHA-256 hash of the data is stored on the Ethereum blockchain to ensure security and immutability. During verification, hashes are compared to determine authenticity.

Significance and Relevance of the Research:

This research addresses the growing issue of fake academic certificates, which affects institutions, employers, and government organizations. Traditional verification methods are slow and error-prone, creating a need for automated solutions.

The use of AI improves fraud detection, while blockchain ensures secure and tamper-proof storage. The project demonstrates the effective integration of modern technologies, making it highly relevant for real-world applications such as recruitment and academic verification.

Aim and Scope of the Project:

The main aim of this project is to develop a secure and automated system for verifying academic certificates. It focuses on reducing manual effort, improving accuracy, and ensuring data integrity.

The scope includes certificate upload, preprocessing, OCR-based data extraction, AI-based forgery detection, and blockchain storage. The system can be applied in universities, companies, and government organizations and can be extended with features like QR-based verification and mobile applications.

Overview of the Report Structure:

This report is organized to present the project in a structured manner. It begins with the introduction and objectives, followed by a literature survey of existing methods. The system architecture and methodology explain the design and working of the system.

Subsequent sections describe the technologies used, implementation details, testing, and results. The report concludes with advantages, limitations, and future enhancements of the proposed system.

1.2 Background and Motivation

The Academic Records Validator project is situated in the field of Information Security, Document Verification Systems, and Artificial Intelligence. This area focuses on ensuring the authenticity, integrity, and reliability of documents. With the growing use of digital platforms in education and recruitment, the need for secure and automated verification systems has become essential.

Traditionally, academic certificate verification was performed manually by contacting issuing authorities or checking physical documents. Although reliable, this process was slow, labor-intensive, and not suitable for large-scale use. With digitization, centralized databases were introduced, but they are vulnerable to data breaches, manipulation, and single points of failure.

Recent research has explored technologies such as Optical Character Recognition (OCR) for text extraction, Artificial Intelligence (AI) for forgery detection, and blockchain for secure data storage. While these approaches improve efficiency and security, most existing systems focus on individual components rather than providing a fully integrated solution.

There is still a gap in developing a system that combines automated data extraction, intelligent fraud detection, and secure storage within a single framework. Many current solutions lack either strong security or effective verification capabilities.

The motivation behind this project is to address these limitations by integrating OCR, AI, and blockchain into a unified system. This approach helps in detecting fake certificates, ensuring data integrity, and automating the verification process. The project is important as it tackles the growing issue of certificate fraud in areas such as education, employment, and governance.

Overall, this work provides a practical and scalable solution for secure academic certificate verification in the modern digital environment.

1.3 Objectives of the Project Work

This section outlines the key goals of the Academic Records Validator project, focusing on developing a secure, efficient, and automated certificate verification system. The objectives aim to address challenges such as certificate forgery, manual verification inefficiencies, and lack of secure data storage.

The main objectives of this project are as follows:

1. To develop an automated system for verifying academic certificates, reducing manual effort and processing time.

2. To implement Optical Character Recognition (OCR) for extracting important details such as name, roll number, institution, and academic information from certificates.

3. To apply image preprocessing techniques using OpenCV to enhance document quality and improve OCR accuracy.

4. To develop an Artificial Intelligence model (MobileNetV2) to detect forged or tampered certificates.
5. To generate a secure hash of certificate data using the SHA-256 algorithm to ensure data integrity.
6. To use blockchain technology (Ethereum) for secure and immutable storage of certificate hash values.
7. To design a user-friendly frontend interface using React for easy certificate upload and verification.
8. To build a backend system using Flask for processing data and integrating system components.
9. To evaluate the system based on accuracy, reliability, and efficiency.
10. To ensure the system is scalable, secure, and suitable for real-world applications.

1.4 Organization of the Report

This section outlines the structure of the report and the flow of content across different chapters. Each chapter focuses on a specific aspect of the Academic Records Validator project, ensuring a logical progression from problem analysis to implementation and evaluation.

- **Chapter 2: Literature Survey**
  This chapter reviews existing work related to certificate verification, OCR, AI-based forgery detection, and blockchain, and identifies the research gap addressed by this project.
- **Chapter 3: System Analysis**
  This chapter explains the limitations of existing systems and defines the requirements of the proposed system.

- **Chapter 4: System Design**

  This chapter describes the system architecture, design diagrams, and interaction between different modules.

- **Chapter 5: System Implementation**

  This chapter presents the development of the system and the use of technologies such as OpenCV, OCR, AI models, blockchain, Flask, and React.

- **Chapter 6: System Testing**

  This chapter covers testing methods used to ensure system accuracy, reliability, and performance.

- **Chapter 7: Results and Discussion**

  This chapter analyzes the results and evaluates the effectiveness of the system in certificate verification.

- **Chapter 8: Conclusions and Future Scope**

  This chapter summarizes the work and suggests future improvements and enhancements.

CHAPTER 2

Literature Survey

2.1 Introduction

The Literature Survey chapter provides a comprehensive review of existing research and technologies related to academic certificate verification systems. The purpose of this section is to analyze previous work carried out in the domains of document verification, Optical Character Recognition (OCR), Artificial Intelligence (AI)-based forgery detection, and blockchain technology. By examining these studies, this chapter establishes a strong foundation for understanding the current state of research and identifies the gaps that the proposed project aims to address.

The literature survey is conducted by reviewing various research papers, journals, technical articles, and existing systems relevant to the problem domain. The methodology involves analyzing the approaches used in these works, comparing their advantages and limitations, and understanding how different technologies have been applied to solve similar problems. Special attention is given to systems that focus on automation, security, and accuracy in certificate verification.

Reviewing prior research is essential as it helps in identifying the strengths and weaknesses of existing solutions. Many traditional systems rely on manual verification or centralized databases, which are often inefficient and vulnerable to manipulation. Recent advancements have introduced the use of OCR for extracting textual information, AI models for detecting forgery, and blockchain for secure data storage. However, most of these approaches are implemented independently and lack a fully integrated solution.

This chapter highlights the relevance of combining multiple modern technologies to create a more robust and efficient system. It provides insights into how previous works have contributed to the field and how the proposed Academic Records Validator builds upon these contributions to offer an improved solution.

The remainder of this chapter is organized as follows: it begins with a review of existing certificate verification methods, followed by studies on OCR-based text extraction, AI-based forgery detection techniques, and the application of blockchain in secure data management. Finally, a summary of the literature is presented, emphasizing the research gap and the need for the proposed system.

2.2 Review of Prior Research

This section presents a comprehensive review of existing research and technologies related to academic certificate verification, document authentication, and fraud detection. The review is organized into key thematic areas, including traditional verification systems, OCR-based text extraction, AI-based forgery detection, and blockchain-based security solutions. This structured approach helps in understanding the evolution of methods and identifying gaps in current systems.

1. Traditional Certificate Verification Systems:
Earlier approaches to certificate verification primarily relied on manual processes, where institutions verified documents by contacting issuing authorities or checking physical copies. While these methods ensured a certain level of authenticity, they were time-consuming, labor-intensive, and not scalable. Later, centralized digital databases were introduced to store and verify academic records. However, such systems suffer from issues like data tampering, lack of transparency, and single points of failure.

2. OCR-Based Document Processing:
Optical Character Recognition (OCR) has been widely used for extracting textual information from scanned documents and images. Tools such as Tesseract OCR have shown significant effectiveness in converting image-based text into machine-readable format. Research in this area focuses on improving accuracy through preprocessing techniques like noise removal, image enhancement, and segmentation. Despite its usefulness, OCR performance can be affected by

poor image quality and complex document layouts.

3. AI-Based Forgery Detection:

Recent advancements in Artificial Intelligence, particularly deep learning, have enabled automated detection of forged documents. Convolutional Neural Networks (CNNs), such as MobileNetV2, are commonly used to analyze visual patterns and identify inconsistencies in images. Studies have demonstrated that AI models can effectively detect tampering, alterations, and fake patterns in documents. However, these models require large and diverse datasets for training and may face challenges in generalizing to unseen formats.

4. Blockchain for Secure Verification:

Blockchain technology has emerged as a promising solution for secure and tamper-proof data storage. Its decentralized and immutable nature ensures that once data is recorded, it cannot be altered. Several research works propose storing academic credentials or their hashes on blockchain platforms like Ethereum. This approach enhances transparency, security, and trust. However, challenges such as scalability, transaction costs, and integration complexity still exist.

5. Integrated Approaches:

Some recent studies attempt to combine multiple technologies, such as OCR and blockchain or AI and document verification systems. While these approaches show improved performance compared to standalone systems, most of them lack a fully integrated framework that combines OCR, AI-based forgery detection, and blockchain security in a seamless manner.

Summary and Research Gap:

From the above review, it is evident that significant progress has been made in individual areas such as text extraction, fraud detection, and secure storage. However, there is a lack of a unified system that integrates all these technologies into a single, efficient, and scalable solution. The proposed Academic Records Validator addresses this gap by combining OCR for data extraction, AI for forgery detection, and blockchain for secure and immutable storage, thereby providing a comprehensive solution for certificate verification.

2.3 Identified Research Gaps

Based on the review of prior research, it is evident that although significant advancements have been made in the areas of document verification, OCR, Artificial Intelligence, and blockchain technology, several important gaps still exist. Identifying these gaps is essential to justify the need for the proposed Academic Records Validator system.

1. Lack of Integrated Systems:

Most existing solutions focus on a single aspect of certificate verification, such as OCR for text extraction, AI for forgery detection, or blockchain for secure storage. However, there is a lack of a fully integrated system that combines all these technologies into a unified framework. This limits the overall efficiency and reliability of verification processes.

2. Dependence on Manual or Semi-Automated Processes:

Many traditional and even some modern systems still rely partially on manual verification or human intervention. This leads to increased processing time, higher chances of human error, and reduced scalability when handling large volumes of certificates.

3. Limitations of OCR Accuracy:

Although OCR technology has improved significantly, it still faces challenges when dealing with low-quality images, complex certificate layouts, or varying fonts and formats. Existing systems often do not incorporate robust preprocessing or error-handling mechanisms to improve OCR accuracy.

4. Insufficient Forgery Detection Mechanisms:

While AI-based approaches have been proposed for detecting forged documents, many systems either do not include such mechanisms or rely on basic rule-based checks. Even in AI-based systems, there is often a lack of comprehensive models trained on diverse datasets, limiting their ability to detect sophisticated forgeries.

5. Security and Data Integrity Issues:

Centralized databases used in many existing systems are vulnerable to data breaches, unauthorized modifications, and single points of failure. Although blockchain has been introduced as a solution, its adoption in certificate verification systems is still limited and not widely implemented in practical applications.

6. Lack of Real-Time and Scalable Solutions:
Several existing approaches are not designed to handle large-scale, real-time verification scenarios. As the number of digital certificates increases, there is a need for systems that can efficiently process and verify data at scale without compromising performance.

7. Limited Focus on Privacy:
Some systems store complete certificate data, raising concerns about data privacy and misuse. There is a need for solutions that ensure verification without exposing sensitive information.

Conclusion of Research Gaps:
The above gaps highlight the need for a comprehensive, automated, and secure solution for academic certificate verification. The proposed project addresses these limitations by integrating OCR for data extraction, AI for intelligent forgery detection, and blockchain for secure and immutable storage, while also ensuring scalability, efficiency, and data privacy.

2.4 Summary

This chapter presented a comprehensive review of existing research related to academic certificate verification systems, focusing on key areas such as traditional verification methods, OCR-based text extraction, AI-based forgery detection, and blockchain technology for secure data storage. The literature highlights that while significant progress has been made in each of these domains, most existing approaches address these components independently rather than as a unified system.

Traditional verification methods were found to be time-consuming and inefficient, while centralized digital systems suffer from security vulnerabilities and lack of transparency. OCR-based techniques have improved the automation of data extraction but still face challenges in accuracy under varying conditions. Similarly, AI-based approaches have shown promising results in detecting forged documents, yet they require robust training and are not widely integrated into practical verification systems. Blockchain technology, on the other hand, offers strong security and immutability but is still underutilized in real-world certificate verification applications.

The analysis of prior research also revealed several critical gaps, including the lack of integrated solutions, dependence on manual processes, limitations in OCR accuracy, insufficient forgery detection mechanisms, and concerns related to security, scalability, and data privacy. These gaps indicate the need for a more comprehensive and efficient system that combines the strengths of multiple technologies.

In response to these challenges, the proposed Academic Records Validator aims to integrate OCR, Artificial Intelligence, and blockchain into a single framework to provide an automated, secure, and reliable certificate verification system. This approach not only addresses the identified limitations but also contributes to advancing the field by offering a practical and scalable solution.

Overall, this literature survey establishes the foundation for the proposed work and clearly demonstrates the necessity of the research. It also provides direction for the subsequent chapters, which focus on the design, implementation, and evaluation of the proposed system.

CHAPTER 3

System Analysis

3.1 Existing System

The existing system for academic certificate verification primarily relies on traditional and semi-digital methods. In most cases, verification is performed manually by institutions or organizations by checking physical documents or contacting the issuing authority. This process involves significant human effort and is often time-consuming, especially when handling a large number of certificates.

With the advancement of technology, some organizations have adopted centralized digital databases to store and verify academic records. In such systems, certificate details are stored in a central server, and verification is performed by matching the provided data with the stored records. While this approach improves speed compared to manual methods, it still has several limitations related to security, scalability, and reliability.

Additionally, many existing systems do not incorporate advanced technologies such as Artificial Intelligence or Blockchain. As a result, they are unable to detect sophisticated forgeries or ensure tamper-proof storage of data. These limitations make the current systems less effective in addressing the growing problem of fake and manipulated certificates.

Overall, the existing systems lack automation, intelligent fraud detection, and strong security mechanisms, which highlights the need for a more advanced and integrated solution.

3.1.1 Disadvantages of Existing System

The existing systems for academic certificate verification suffer from several limitations that affect their efficiency, reliability, and security. These drawbacks highlight the need for an improved and automated solution.

- **Time-Consuming Process:** Manual verification requires contacting institutions or checking physical records, which takes a significant amount of time.
- **Human Errors:** Since the process involves human intervention, there is a high possibility of mistakes during verification.
- **Lack of Scalability:** Existing systems are not efficient in handling large volumes of certificates, especially during mass recruitment or admissions.
- **Security Vulnerabilities:** Centralized databases are prone to data breaches, unauthorized access, and manipulation.
- **No Forgery Detection:** Traditional systems lack intelligent mechanisms to detect tampered or fake certificates.
- **Lack of Transparency:** Verification processes are not always transparent, making it difficult to track or audit the authenticity of records.
- **Dependency on Third Parties:** Verification often depends on issuing authorities, causing delays and inefficiencies.

3.2 Proposed System

The proposed system, **Academic Records Validator**, is an advanced and automated solution designed to overcome the limitations of existing systems. It integrates modern technologies such as Optical Character Recognition (OCR), Artificial Intelligence (AI), and Blockchain to provide a secure, efficient, and reliable certificate verification process.

In this system, users can upload academic certificates in the form of images or PDF documents. The system processes these documents using image preprocessing techniques to enhance quality and improve accuracy. It then uses OCR to extract important information such as name, roll number, institution, and academic details.

To ensure authenticity, an AI-based model (MobileNetV2) analyzes the certificate to detect any signs of forgery or tampering. Additionally, a unique hash of the extracted data is generated using the SHA-256 algorithm. This hash is stored securely on the blockchain using smart contracts, ensuring that the data remains immutable and tamper-proof.

During verification, the system compares the newly generated hash with the stored hash on the blockchain. If both match, the certificate is considered genuine; otherwise, it is flagged as fake. This process ensures both accuracy and security.

Overall, the proposed system provides a fast, automated, and highly secure solution for academic certificate verification, addressing the shortcomings of traditional methods.

3.2.1 Advantages of Proposed System

The proposed Academic Records Validator system offers several advantages over traditional verification methods by leveraging modern technologies such as OCR, Artificial Intelligence, and Blockchain.

- **Automation:** The entire verification process is automated, reducing the need for manual intervention and saving time.
- **High Accuracy:** OCR and AI models ensure accurate data extraction and reliable detection of forged certificates.
- **Enhanced Security:** Blockchain technology provides tamper-proof storage, ensuring data integrity and preventing unauthorized modifications.
- **Fast Processing:** The system significantly reduces verification time compared to traditional methods.
- **Scalability:** It can efficiently handle large volumes of certificates, making it suitable for real-world applications.
- **Transparency:** Blockchain ensures transparency and traceability in the verification process.
- **Reduced Human Errors:** Automation minimizes the chances of mistakes caused by manual verification.

- **Data Privacy:** Only hashed data is stored on the blockchain, protecting sensitive certificate information.

3.3 System Requirements

This section describes the hardware and software requirements necessary for the development and deployment of the proposed Academic Records Validator system.

Hardware Requirements

- **Processor:** Minimum Intel Core i3; Recommended Intel Core i5/i7 or AMD Ryzen 5/7
- **RAM:** Minimum 8 GB; Recommended 16 GB
- **Storage:** Minimum 10 GB free space; Recommended 20+ GB (SSD preferred)
- **Internet Connectivity:** Required for dependency installation and blockchain interactions
- **GPU (Optional):** NVIDIA GPU with CUDA support for faster AI model processing (MobileNetV2)

Software Requirements

- **Operating System:** Windows, Linux, or macOS
- **Programming Languages:** Python, JavaScript
- **Frontend:** React.js
- **Backend Framework:** Flask (Python)
- **OCR & Image Processing:** Tesseract OCR, OpenCV, Pillow (PIL)
- **Artificial Intelligence:** TensorFlow (MobileNetV2)
- **Blockchain Platform:** Ethereum

- **Smart Contract Language:** Solidity
- **Blockchain Tools:** Hardhat, Web3.py
- **Libraries:** NumPy, Pandas, Axios
- **Development Tools:** Visual Studio Code / PyCharm
- **Browser:** Google Chrome

The Academic Records Validator is a modern system designed to automate the process of verifying academic certificates and eliminate the use of fake or tampered documents. In traditional systems, certificate verification is performed manually, which is time-consuming and prone to errors. This project introduces a digital solution that ensures fast, accurate, and secure verification using advanced technologies.

The system begins with the user uploading an academic certificate in the form of an image or PDF. The uploaded document is processed using image preprocessing techniques such as noise removal, grayscale conversion, and image enhancement. These steps improve the clarity of the certificate and prepare it for further processing.

After preprocessing, Optical Character Recognition (OCR) is used to extract important information from the certificate. The extracted details include the student's name, roll number, institution name, year of passing, and academic performance. This data is then structured into a proper format for further analysis.

To enhance security, an Artificial Intelligence model based on MobileNetV2 is used to detect forgery. The model analyzes the visual features of the certificate and identifies any inconsistencies, alterations, or suspicious patterns. This helps in detecting fake certificates that may appear similar to genuine ones.

Once the data is extracted, a unique digital fingerprint is generated using the SHA-256 hashing algorithm. This hash value represents the certificate data and is highly sensitive to any changes. Even a small modification in the data results in a completely different hash value.

The generated hash is stored on the Ethereum blockchain using smart contracts. Blockchain ensures that the stored data is immutable, secure, and transparent. Instead of storing the entire certificate, only the hash is stored, which protects user privacy while ensuring data integrity.

During verification, the certificate is uploaded again, and the same process is repeated to generate a new hash. This hash is compared with the one stored on the blockchain. If both hashes match, the certificate is verified as genuine; otherwise, it is identified as fake.

Here is a table representing a simple example of structured data:

Field	**Example Value**
Name	John Doe
Roll Number	12345
Institution	XYZ University
CGPA	8.5

Table 3.1: Sample Extracted Certificate Data

Overall, the proposed system provides a reliable, efficient, and secure method for academic certificate verification. By integrating OCR, Artificial Intelligence, and Blockchain technology, it reduces manual effort, improves accuracy, and ensures trust in the verification process.

CHAPTER 4

System Design

4.1 System Architecture

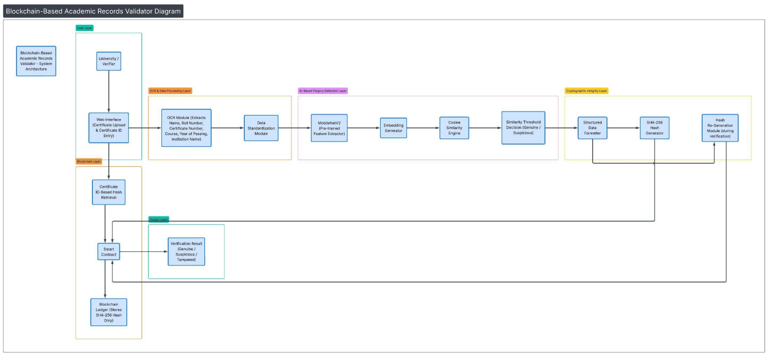

Figure 4.1: System Architecture Diagram.

4.2 Data Flow Diagram

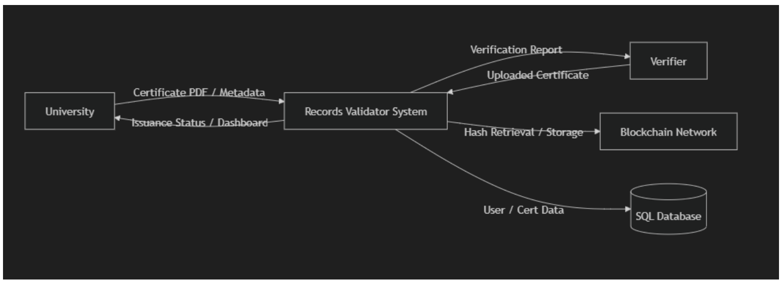

Figure 4.2: Data Flow Diagram

4.3 UML Diagrams

4.3.1 Class Diagram

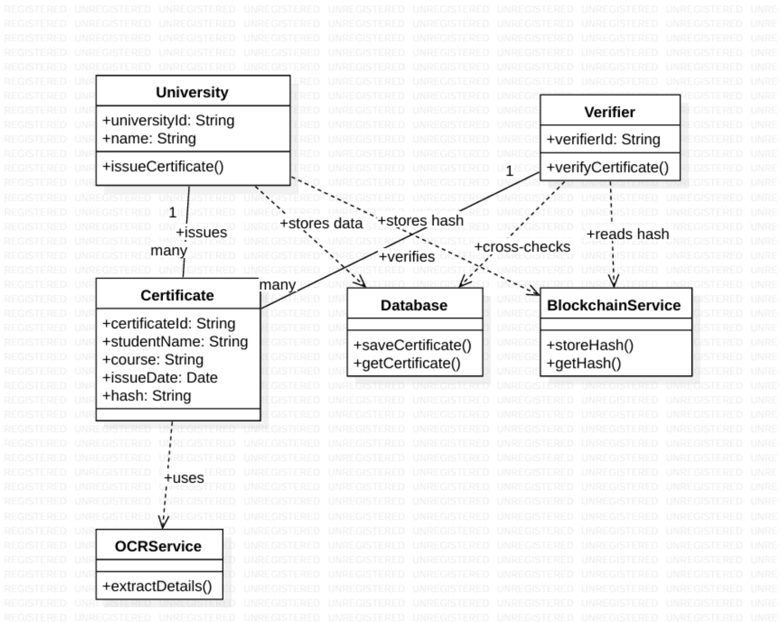

Figure 4.3: Class Diagram

4.3.2 Object Diagram

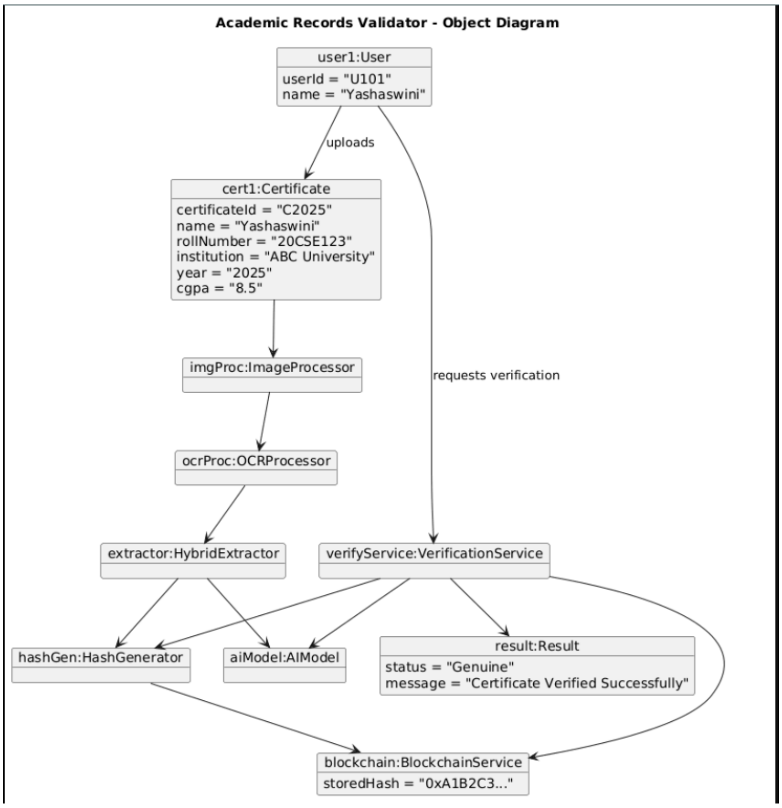

Figure 4.4: Object Diagram

4.3.3 Collaboration Diagram

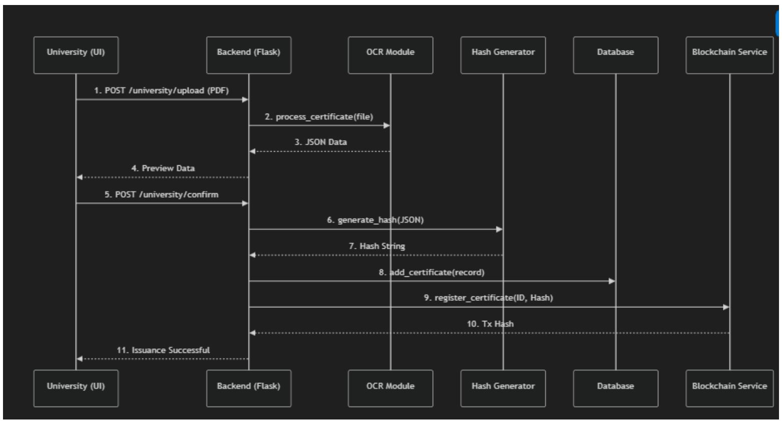

Figure 4.5: Collaboration Diagram

4.3.4 Use Case Diagram

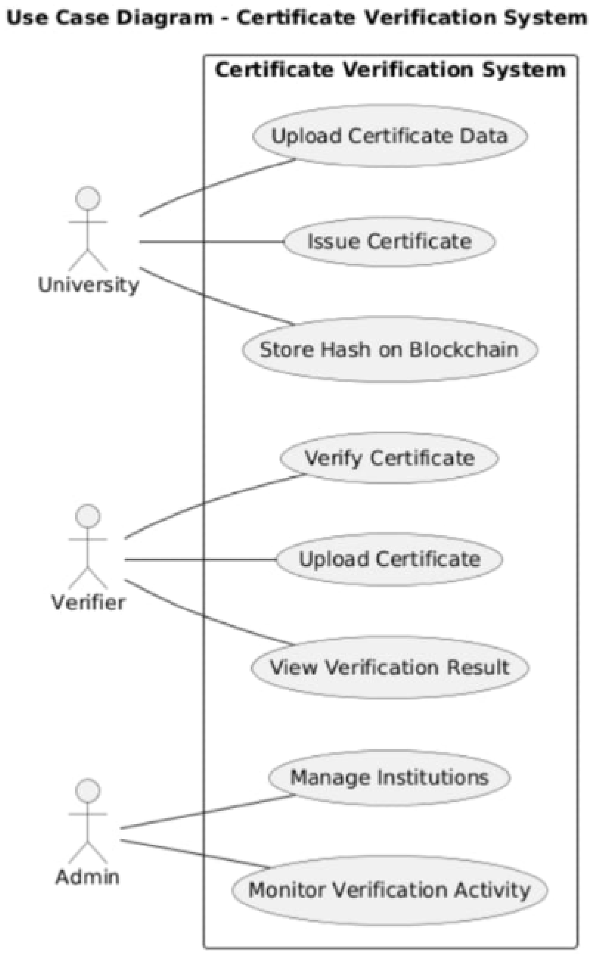

Figure 4.6: Use Case Diagram

4.3.5 Sequence Diagram

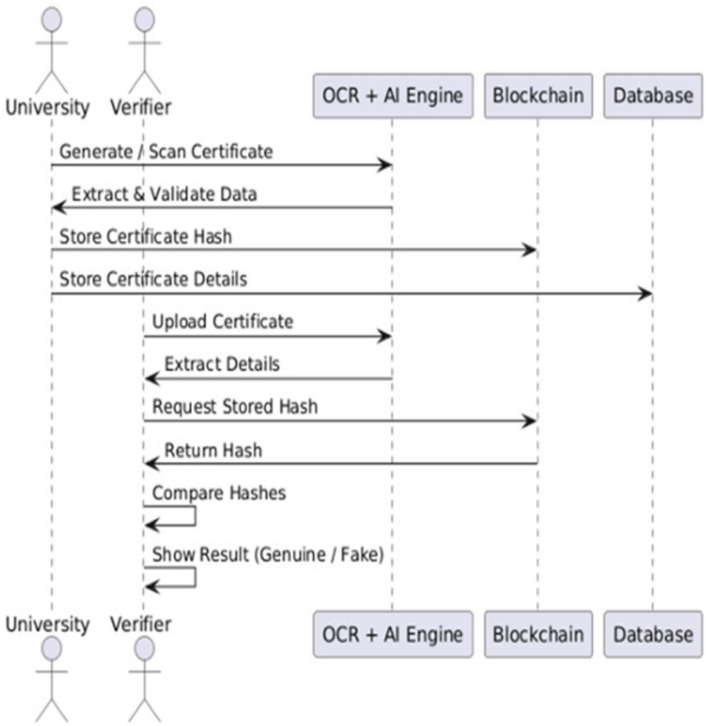

Figure 4.7: Sequence Diagram

4.3.6 Activity Diagram

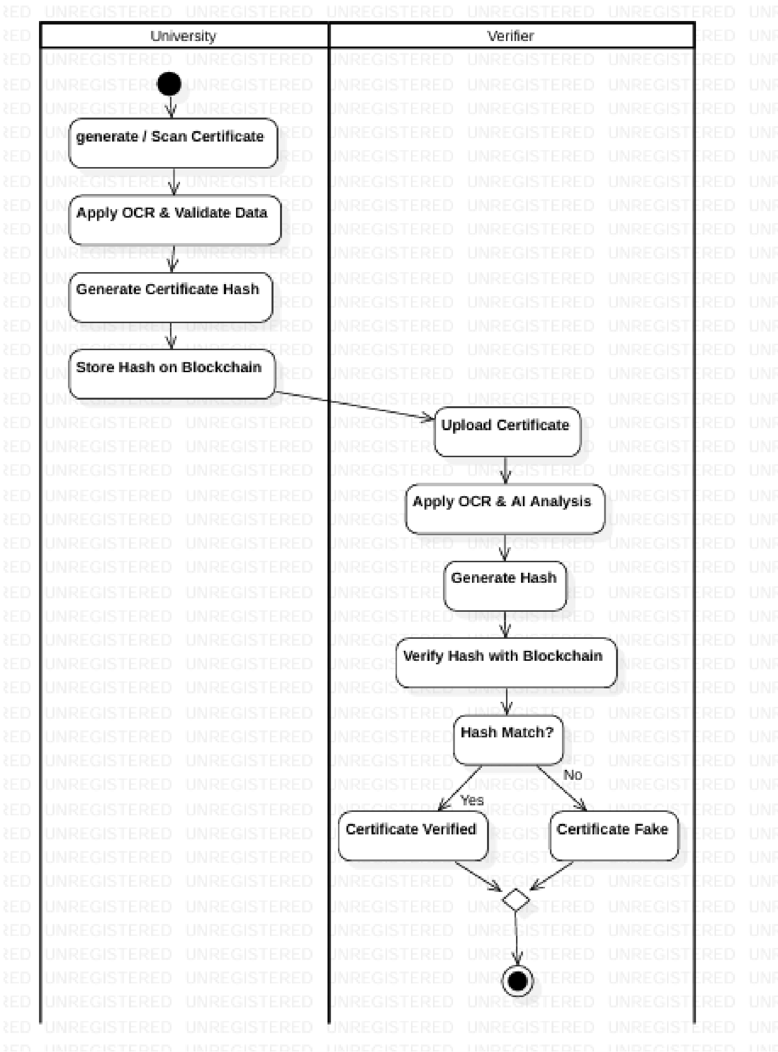

Figure 4.8: Activity Diagram

4.3.7 Component Diagram

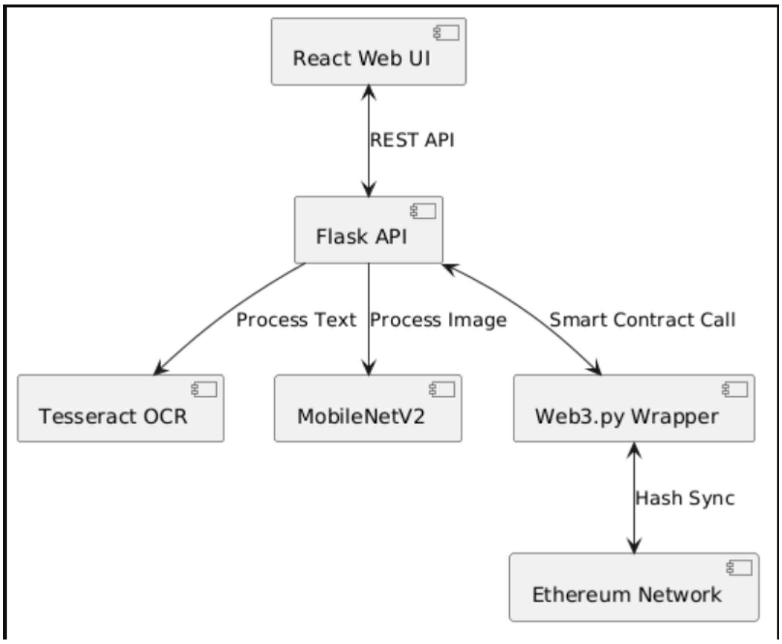

Figure 4.9: Component Diagram

4.3.8 Deployment Diagram

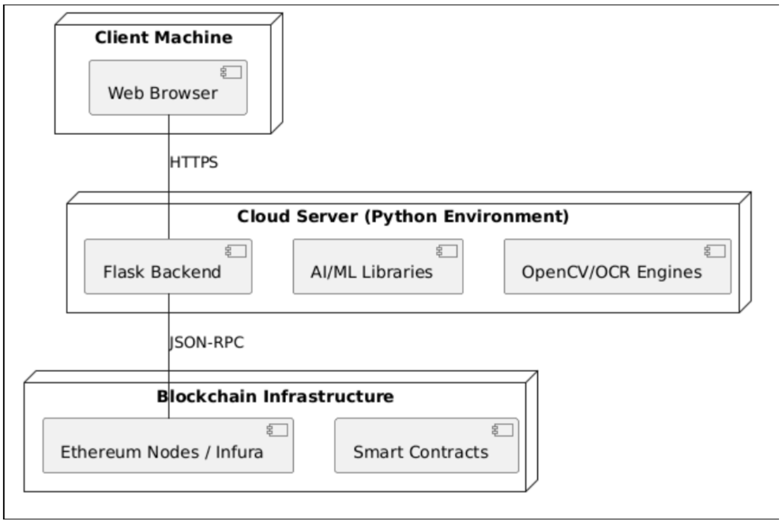

Figure 4.10: Deployment Diagram

CHAPTER 5

System Implementation

5.1 Modules Description

The Academic Records Validator system is divided into several functional modules to ensure efficient processing and verification of academic certificates.

- **User Module:**
  This module allows users to upload academic certificates and request verification. It provides an interface to view the verification results.

- **Image Processing Module:**
  This module enhances the quality of uploaded certificates using techniques such as grayscale conversion, noise removal, and image sharpening to improve accuracy.

- **OCR Module:**
  The Optical Character Recognition (OCR) module uses Tesseract to extract important details such as name, roll number, certificate ID, institution name, year of passing, and marks from the certificate.

- **Data Extraction Module (Hybrid Extractor):**
  This module processes the raw OCR output and organizes it into structured fields for further processing.

- **AI-Based Forgery Detection Module:**
  This module uses MobileNetV2 to analyze certificate images and detect whether they are genuine or fake based on visual patterns and inconsistencies.

- **Hash Generation Module:**
  This module generates a unique SHA-256 hash based on the extracted certificate data to ensure data integrity.

- **Blockchain Module:**

  This module stores and retrieves certificate hashes on the Ethereum blockchain using smart contracts, ensuring immutability and security.

- **Verification Module:**

  This module compares the newly generated hash with the stored blockchain hash and combines AI results to determine whether the certificate is genuine or fake.

5.2 Technology Used

The Academic Records Validator project is developed using a combination of modern technologies to ensure accurate, secure, and efficient certificate verification.

Frontend:

React.js is used to build an interactive and user-friendly interface for uploading certificates and displaying results.

Backend:

Flask (Python) is used to handle API requests, process certificate data, and manage communication between different modules.

OCR Technology:

Tesseract OCR is used to extract text from certificate images and PDFs.

Image Processing:

OpenCV and Pillow (PIL) are used to enhance image quality through preprocessing techniques.

Artificial Intelligence:

TensorFlow with MobileNetV2 is used for detecting forged or manipulated certificates based on visual patterns.

Blockchain Technology:

Ethereum blockchain is used for secure and tamper-proof storage of certificate hashes using Solidity smart contracts and Hardhat environment.

Hashing Algorithm:

SHA-256 is used to generate a unique digital fingerprint for each certificate.

Libraries & Tools:
NumPy, Pandas (data processing), Web3.py (blockchain interaction), and Axios (API communication).

5.3 Algorithms used

Input: Certificate image/PDF
Output: Verification result (Genuine / Fake)

1. Start
2. Upload the academic certificate (10th / Intermediate / Degree)
3. Apply image preprocessing:
   - Convert to grayscale
   - Remove noise
   - Enhance image quality
4. Perform OCR using Tesseract:
   - Extract text from certificate
5. Apply Hybrid Extraction:
   - Identify and structure fields (name, roll number, certificate ID, etc.)
6. Perform AI-based forgery detection:
   - Use MobileNetV2 to check if the certificate visually appears genuine or fake
7. Generate SHA-256 hash:
   - Create a unique hash from extracted data
8. Store hash on blockchain (during registration phase)
9. For verification:

- Re-extract data from uploaded certificate
- Generate new SHA-256 hash

10. Retrieve stored hash from blockchain
11. Compare hashes:
    - If both hashes match → Certificate is Genuine
    - Else → Certificate is Fake
12. Combine AI result with hash comparison for final decision
13. Display verification result
14. End

5.4 Coding

```
import React, { useState } from 'react';
import { universityAPI } from '../services/api';

const UniversityReview = ({ extractedData, onConfirmSuccess, onBack }) => {

    const [academicData, setAcademicData] = useState({
     tenth_certificate: extractedData.academic_data?.tenth_certificate || {
          name: extractedData.student_name || '',
          certificate_number: extractedData.certificate_id || '',
          roll_number: extractedData.roll_number || '',
          institution_name: extractedData.university || '',
          year_of_passing: extractedData.year || '',
          course_or_stream: 'SSC',
          cgpa_or_marks: extractedData.cgpa || ''
      },
     intermediate_certificate: extractedData.academic_data?.intermediate_certifi
          name: '', certificate_number: '', roll_number: '',
```

```
        institution_name: '', year_of_passing: '',
        course_or_stream: '', cgpa_or_marks: ''
    },
  degree_certificate: extractedData.academic_data?.degree_certificate || {
        name: '', certificate_number: '', roll_number: '',
        institution_name: '', year_of_passing: '',
        course_or_stream: '', cgpa_or_marks: ''
    }
});

const [loading, setLoading] = useState(false);
const [error, setError] = useState('');

const handleFieldChange = (certType, field, value) => {
    setAcademicData({
        ...academicData,
        [certType]: {
            ...academicData[certType],
            [field]: value
        }
    });
};

const handleConfirm = async () => {
    setError('');

    if (!academicData.tenth_certificate.certificate_number &&
    !academicData.intermediate_certificate.certificate_number &&
        !academicData.degree_certificate.certificate_number) {
    setError('Please provide at least one Certificate ID for any level.');
        return;
    }
```

```
        setLoading(true);

        try {
            const result = await universityAPI.confirmCertificate({
                ...academicData.degree_certificate,
                academic_data: academicData
            });

            if (result.success) {
                onConfirmSuccess(result.data);
            } else {
                setError(result.error || 'Confirmation failed');
            }
        } catch (err) {
            setError('System error during confirmation.');
        } finally {
            setLoading(false);
        }
    };

    return (
        <div>
            {/* UI rendering code */}
        </div>
    );
};

export default UniversityReview;
```

CHAPTER 6

System Testing

6.1 Testing Methodology

The Academic Records Validator system is tested using both functional and non-functional testing methods to ensure accuracy, reliability, and performance. Unit testing is performed on individual modules such as OCR, AI model, and blockchain interaction. Integration testing ensures smooth communication between modules.

The system is also tested for different inputs, including valid and invalid certificates, to check error handling and robustness. Overall, the testing ensures that the system provides correct verification results and performs efficiently.

6.2 Testing under Various Stages

- **Unit Testing:**
  Each module such as OCR, image processing, AI model, and blockchain is tested individually.

- **Integration Testing:**
  The interaction between modules (OCR, AI, hashing, blockchain) is tested to ensure proper data flow.

- **System Testing:**
  The complete system is tested with real certificate inputs to verify end-to-end functionality.

- **Performance Testing:**
  The system is tested for speed and response time during certificate processing and verification.

- **User Acceptance Testing:**

  The system is evaluated from a user perspective to ensure ease of use and correct output.

6.3 Test Cases

S.No	Test Case	Expected Output	Result
1	Upload valid certificate	Certificate processed successfully	Pass
2	Upload invalid certificate	Error message displayed	Pass
3	OCR extraction	Correct extraction of details	Pass
4	AI forgery detection	Fake certificate detected	Pass
5	Hash generation	Unique SHA-256 hash generated	Pass
6	Blockchain storage	Hash stored successfully	Pass
7	Verification (valid)	Output: Genuine	Pass
8	Verification (fake)	Output: Fake	Pass

Table 6.1: Test Cases for Academic Records Validator

CHAPTER 7

Results and Discussion

7.1 Introduction

This chapter presents the results obtained from the Academic Records Validator system and discusses their significance. The results demonstrate how effectively the system verifies academic certificates using OCR, AI-based forgery detection, and blockchain technology.

The purpose of this chapter is to analyze the performance of the system in terms of accuracy, reliability, and efficiency. It highlights how the proposed system meets the objectives of secure and automated certificate verification.

The chapter is organized into different sections, including an overview of results, result screenshots, analysis and interpretation, and a summary of findings.

7.2 Overview of Results

The Academic Records Validator system successfully performs certificate verification through a combination of OCR, AI, and blockchain technologies. The system accurately extracts relevant details such as name, roll number, certificate ID, and institution name from the uploaded certificates.

The AI-based MobileNetV2 model effectively identifies forged or manipulated certificates by analyzing visual patterns. The SHA-256 hashing algorithm generates unique hashes for each certificate, and the blockchain ensures secure and tamper-proof storage of these hashes.The system demonstrates high accuracy in verifying genuine certificates and correctly identifying fake ones. It also reduces manual effort and processing time, making the verification process faster and more efficient.

Overall, the results show that the system is reliable, secure, and suitable for real-world applications.

7.2.1 Result Screenshots

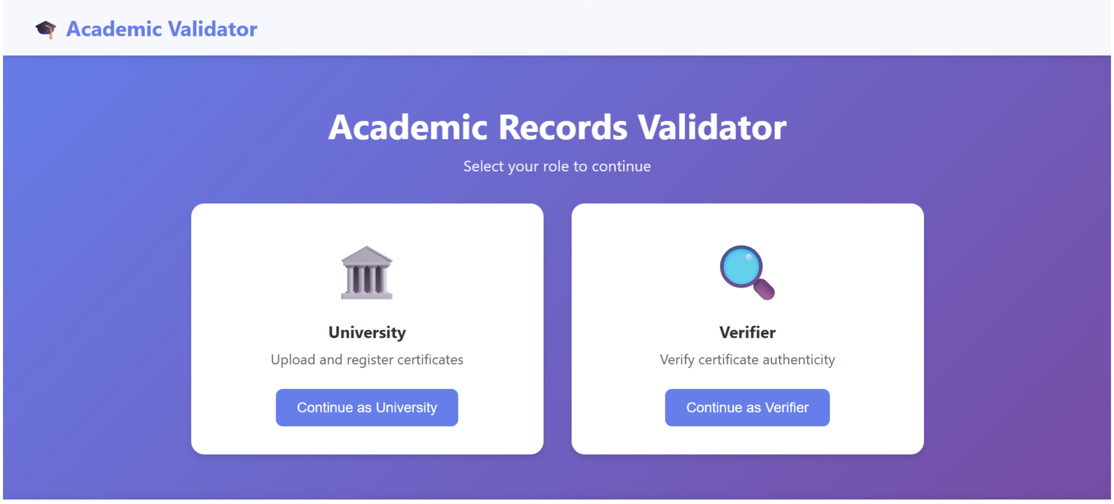

Figure 7.1: Dashboard

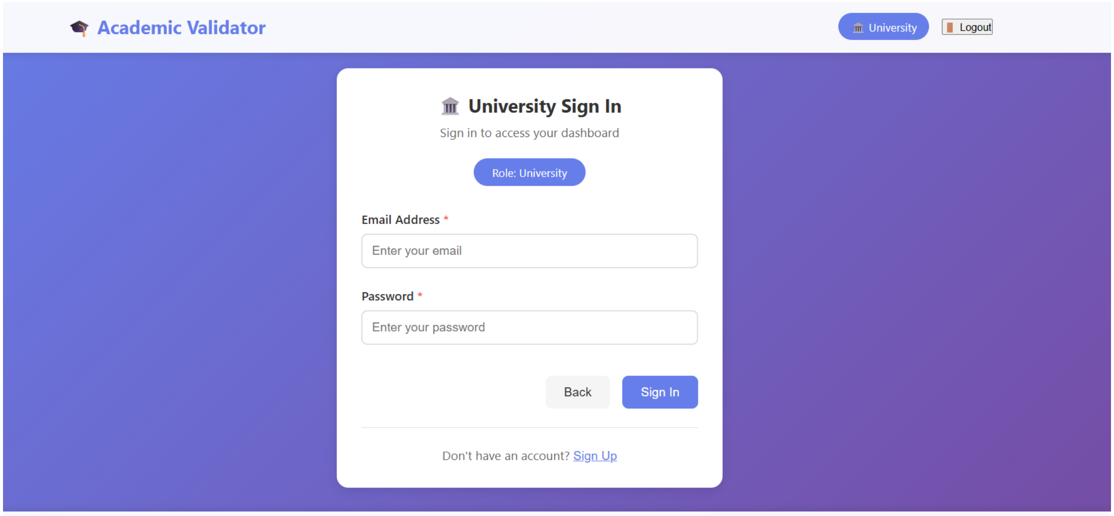

Figure 7.2: University Login

7.2.2 Summary

In this chapter, the results of the Academic Records Validator system were presented and analyzed. The system demonstrated its ability to accurately extract data, detect forgery using AI, and verify certificates using blockchain technology.

The findings confirm that the proposed system provides a secure, reliable, and efficient solution for academic certificate verification. It reduces manual effort, improves accuracy, and enhances trust in the verification process.

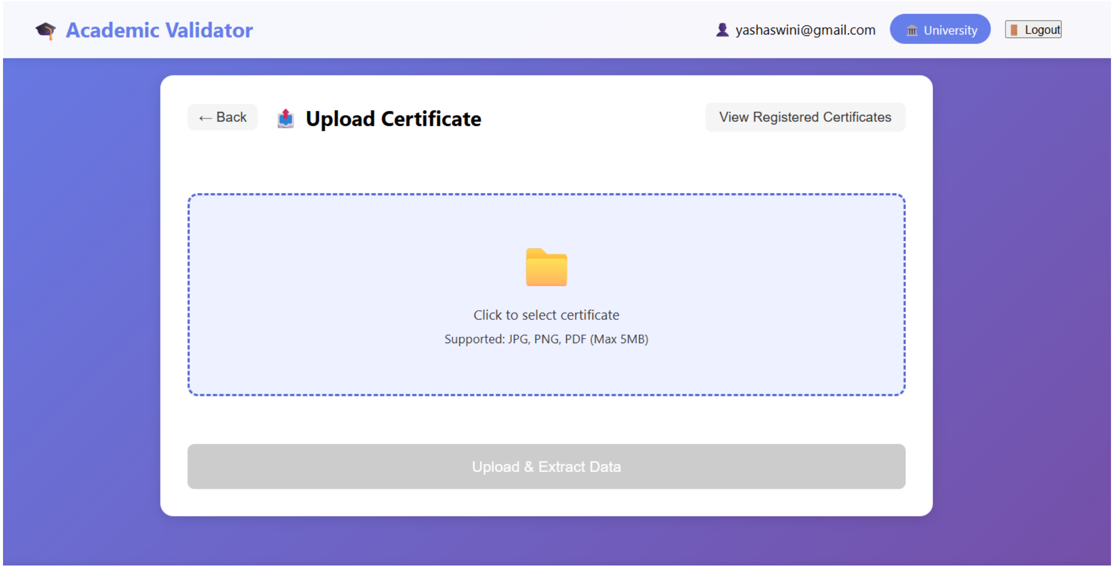

Figure 7.3: University Portal

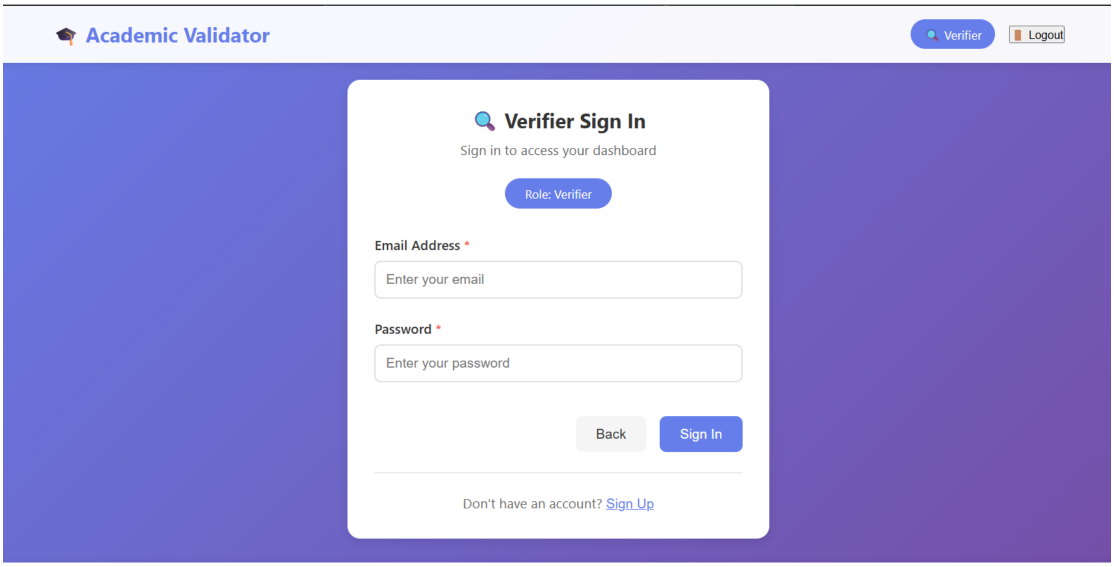

Figure 7.4: Verifier Login

Thus, the results validate the effectiveness of the system and its potential for real-world implementation.

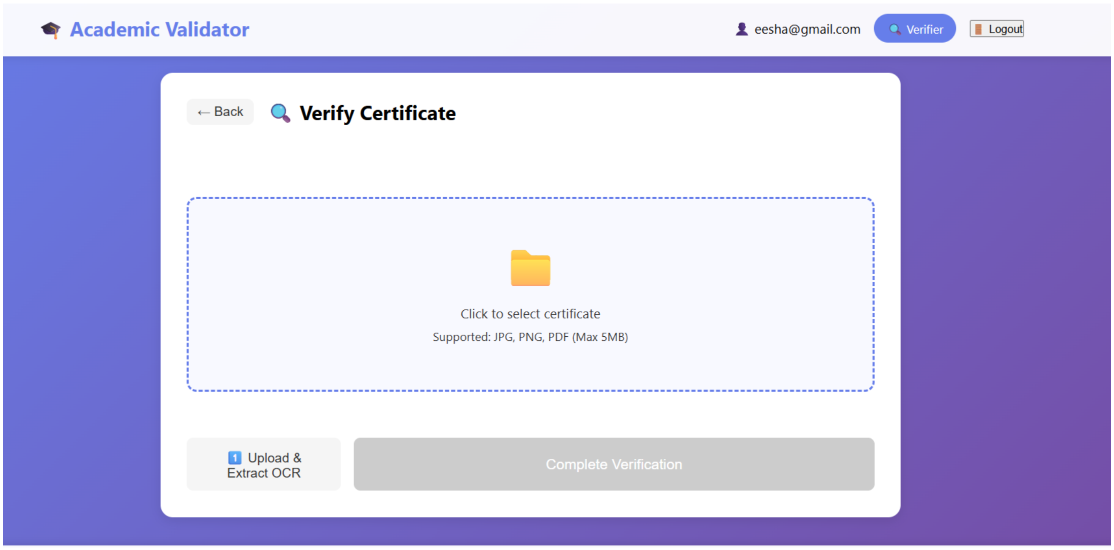

Figure 7.5: Verifier Portal

CHAPTER 8

Conclusions and Future Scope

8.1 Conclusions

The Academic Records Validator project successfully demonstrates a secure and automated approach for verifying academic certificates using modern technologies such as OCR, Artificial Intelligence, and Blockchain.The system effectively extracts certificate data using OCR, detects forgery using the MobileNetV2 model, and ensures data integrity through SHA-256 hashing and blockchain storage. The verification process is fast, reliable, and reduces manual effort significantly.

The project meets its primary objectives of providing a tamper-proof, efficient, and scalable solution for certificate verification. It enhances transparency and trust in academic credential validation and can be effectively used in real-world applications such as universities, organizations, and recruitment systems.

Overall, the project proves that integrating AI and blockchain technologies can greatly improve the accuracy and security of document verification systems.

8.2 Future Scope of Work

Although the Academic Records Validator system provides an effective solution, there are several opportunities for further improvement and enhancement.

The system can be extended to support a wider range of documents such as diplomas, professional certifications, and government-issued documents. Advanced AI models can be implemented to improve forgery detection accuracy and handle more complex manipulations.Integration with national or global academic databases can further strengthen the verification process by enabling real-time validation from official sources. Additionally, deploying the system on cloud platforms can improve scalability and accessibility.

The OCR accuracy can be enhanced by training custom models for different certificate formats. Multi-language support can also be added to handle certificates in regional languages.Future work may also include developing a mobile application for easy access and improving the user interface for better usability. Addressing these aspects will make the system more robust, efficient, and suitable for large-scale deployment.

Thus, the project provides a strong foundation for further research and development in secure document verification systems.

This section discusses the potential directions for future research or work that can build upon your project. It explores areas that require further investigation, additional experiments, or improvements. You should suggest topics or questions that remain unanswered, new areas of application, or possible extensions of the current work. This shows the ongoing relevance of the project and provides a roadmap for future research or practical implementation.

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